

Supplementary Methods, Diagnostics and Replication Inventory

Supplementary Information for the anonymous manuscript “Regulatory Resilience in Technology Transfer Systems”.

S1 Corpus construction and qualitative coding

The corpus contains 34 Vietnamese-language files: 27 news or policy documents and 7 legal or professional documents. Each file is treated as one document. The inventory supplied with the replication package provides the anonymous document identifier, filename, operational genre flag and dual-role-source flag. Full texts are withheld from the public package because they reproduce third-party publications.

First-cycle coding recorded the transfer object, actor, reported problem, institutional level and legal interface. Deductive sensitising codes covered performance, adaptation, information, regulatory coordination and complementary capability. Inductive object-specific codes covered acceptance, tacit know-how, restrictive licensing, data access, AI roles, valuation, semiconductor capability and agricultural diffusion. Second-cycle comparison grouped passages into mechanisms only when the functional distinction changed the relevant legal or policy response. The main manuscript's evidence matrix records representative traces, bounded inferences and limiting cases. The corpus does not support claims of prevalence, theoretical saturation or inter-coder reliability.

Preprocessing lower-cases text, removes URLs, expands selected unambiguous abbreviations, uses lexicon-based maximum-matching tokenisation with domain multi-word terms, and removes general and structural stop words. TF-IDF uses $\min_df=2$ and $\max_df=0.9$. NMF uses `nndsvda` initialization, seed 42 and 500 maximum iterations. K-means uses the same TF-IDF matrix, seed 42 and ten initializations.

S2 Supplementary text-model sensitivity audit

The following diagnostics are not treated as statistical evidence and do not validate the qualitative coding. They are reported solely to make genre sensitivity visible in the bounded corpus. $K=7$ is retained as the historical specification used for the audit, not as an estimate of a true number of latent topics.

K	Cosine silhouette	Largest topic (%)	ARI NMF–K-means
3	0.243	55.9	0.614
4	0.225	52.9	0.586
5	0.209	44.1	0.836
6	0.183	38.2	0.510
7 retained	0.178	38.2	0.499
8	0.185	38.2	0.740

Table S1 Sensitivity scan on the 34-document corpus

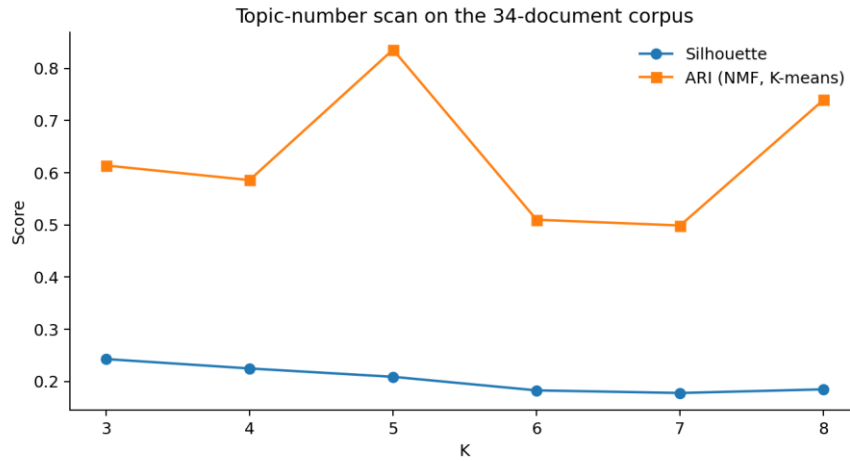


Fig. S1 Silhouette and ARI across $K = 3-8$

S3 Robustness checks

Leave-one-out: after removing Bùi Thị Hằng Nga (2019), a contract-and-dispute topic remains. The other two contract-law documents join that topic with one news item on foreign-investment transfer practice. The distinctive essential-facilities keywords do not recur.

News/policy only: re-estimation on 27 news and policy documents produces themes for general transfer activity, industrial AI and data, agriculture, semiconductors, university innovation, strategic technology or cybersecurity, and mechanical engineering. No contract-law or essential-facilities theme is identified from the reported keywords.

S4 Replication files

File	Purpose
corpus_inventory.csv	Anonymous inventory and genre flags
scripts/run_robustness.py	Documented TF-IDF, NMF, K scan, sensitivity checks and legacy MDS output
results/doc_assignment_k7.csv	Full-corpus $K=7$ assignments
results/doc_assignment_leave_one_out_bui2019.csv	Leave-one-out assignments
results/topics_k7.csv	Full-corpus topic keywords
results/topics_leave_one_out_bui2019.csv	Leave-one-out topic keywords
results/topics_news_policy_only.csv	News/policy-only topic keywords
results/topics_legal_scholarly_only.csv	Legal/professional-only topic keywords
results/k_scan.csv	$K=3-8$ diagnostics
results/fig_mds.png	Legacy diagnostic retained for reproducibility; not used in the revised manuscript

S5 Reproducibility boundary

The public outputs can be inspected without the copyrighted full texts. Exact re-estimation requires access to those texts and the preprocessing lexicons. The package documents the NMF diagnostics and sensitivity checks; it does not reproduce the earlier LDA model, validate the qualitative coding, estimate prevalence, or establish theoretical saturation or inter-coder reliability.